

CUSTOMER SALES ANALYTICS DASHBOARD

END-TO-END BUSINESS INTELLIGENCE SOLUTION USING POWER BI

TECHNOLOGIES USED :-

- POWER BI
- SQL
- DAX
- POWER QUERY
- DATA MODELING

DATE

JULY 2026

PREPARED BY :-

MUSKAN SAINI

B.TECH COMPUTER SCIENCE ENGINEERING

ASPIRING DATA ANALYST

TABLE OF CONTENTS :-

- 1. PROJECT OVERVIEW**
- 2. BUSINESS PROBLEM**
- 3. PROJECT OBJECTIVES**
- 4. DATASET OVERVIEW**
- 5. DATA MODEL**
- 6. DATA CLEANING & TRANSFORMATION**
- 7. DATA MODELING**
- 8. DAX MEASURES**
- 9. EXECUTIVE DASHBOARD**
- 10. SALES DASHBOARD**
- 11. CUSTOMER DASHBOARD**
- 12. CAMPAIGN DASHBOARD**
- 13. INVENTORY DASHBOARD**
- 14. BUSINESS INSIGHTS**
- 15. CHALLENGES**
- 16. FUTURE IMPROVEMENTS**
- 17. SKILLS DEMONSTRATED**
- 18. CONCLUSION**

PROJECT OVERVIEW :-

THIS PROJECT PRESENTS A COMPREHENSIVE BUSINESS INTELLIGENCE SOLUTION DEVELOPED USING MICROSOFT POWER BI. THE OBJECTIVE IS TO TRANSFORM RAW TRANSACTIONAL DATA INTO MEANINGFUL BUSINESS INSIGHTS THROUGH INTERACTIVE DASHBOARDS, EFFICIENT DATA MODELING, AND ADVANCED DAX CALCULATIONS.

THE SOLUTION ANALYZES MULTIPLE BUSINESS AREAS, INCLUDING SALES PERFORMANCE, CUSTOMER BEHAVIOR, MARKETING CAMPAIGN EFFECTIVENESS, AND INVENTORY MANAGEMENT. BY IMPLEMENTING A STAR SCHEMA DATA MODEL AND CREATING INTERACTIVE REPORTS, THE PROJECT ENABLES STAKEHOLDERS TO MONITOR KEY PERFORMANCE INDICATORS, IDENTIFY TRENDS, AND SUPPORT DATA-DRIVEN DECISION-MAKING.

THE REPORT CONSISTS OF FIVE INTERACTIVE DASHBOARDS DESIGNED TO ANSWER REAL-WORLD BUSINESS QUESTIONS AND PROVIDE ACTIONABLE INSIGHTS.

BUSINESS PROBLEM :-

ORGANIZATIONS OFTEN COLLECT LARGE VOLUMES OF OPERATIONAL DATA BUT STRUGGLE TO CONVERT IT INTO MEANINGFUL INSIGHTS. DECISION-MAKERS REQUIRE CENTRALIZED DASHBOARDS THAT PROVIDE A CLEAR OVERVIEW OF BUSINESS PERFORMANCE.

KEY BUSINESS CHALLENGES INCLUDE:

- DIFFICULTY TRACKING SALES PERFORMANCE.**
- LIMITED VISIBILITY INTO CUSTOMER PURCHASING BEHAVIOR.**
- INEFFICIENT EVALUATION OF MARKETING CAMPAIGNS.**
- LACK OF INVENTORY MONITORING.**
- TIME-CONSUMING MANUAL REPORTING.**

THIS PROJECT ADDRESSES THESE CHALLENGES BY CREATING INTERACTIVE POWER BI DASHBOARDS THAT CONSOLIDATE INFORMATION INTO A SINGLE REPORTING SOLUTION.

PROJECT OBJECTIVES :-

THE PRIMARY OBJECTIVES OF THIS PROJECT ARE:

- ANALYZE OVERALL BUSINESS PERFORMANCE.
- IDENTIFY HIGH-PERFORMING PRODUCTS.
- EVALUATE REGIONAL SALES TRENDS.
- UNDERSTAND CUSTOMER PURCHASING PATTERNS.
- MEASURE CAMPAIGN EFFECTIVENESS USING ROI AND CTR.
- MONITOR INVENTORY LEVELS AND INVENTORY VALUE.
- DEVELOP A SCALABLE STAR SCHEMA DATA MODEL.
- CREATE INTERACTIVE DASHBOARDS FOR BUSINESS USERS.



DATASET OVERVIEW :-

THE PROJECT USES A STAR SCHEMA CONSISTING OF MULTIPLE FACT AND DIMENSION TABLES.

FACT TABLES

- FACT_SALES
- FACT_CAMPAGN_SPEND
- FACT_INVENTORY

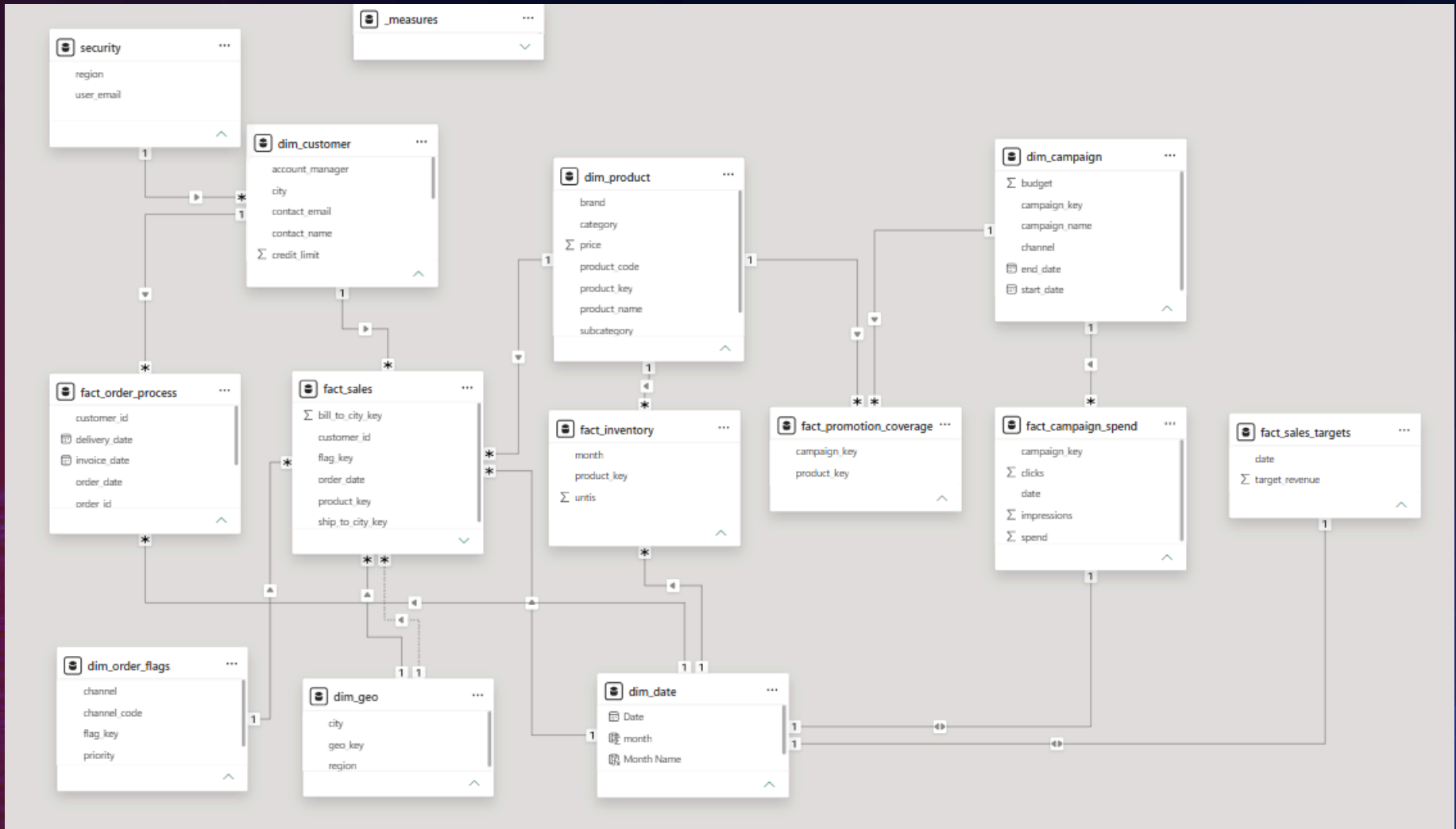
DIMENSION TABLES

- DIM_CUSTOMER
- DIM_PRODUCT
- DIM_CAMPAGN
- DIM_GEO
- DIM_DATE

THE DATASETS CONTAIN INFORMATION RELATED TO SALES TRANSACTIONS, CUSTOMER DETAILS, MARKETING CAMPAIGNS, PRODUCTS, SUPPLIERS, GEOGRAPHY, AND INVENTORY.



DATA MODEL :-



A STAR SCHEMA WAS IMPLEMENTED TO IMPROVE QUERY PERFORMANCE AND SIMPLIFY REPORT DEVELOPMENT. DIMENSION TABLES ARE CONNECTED TO FACT TABLES THROUGH PRIMARY AND FOREIGN KEY RELATIONSHIPS, ENABLING EFFICIENT FILTERING AND DAX CALCULATIONS.



DATA CLEANING & TRANSFORMATION :-

POWER QUERY WAS USED TO PREPARE THE DATA BEFORE BUILDING THE DASHBOARDS.

THE FOLLOWING TRANSFORMATIONS WERE PERFORMED:

- **REMOVED DUPLICATE RECORDS.**
- **CORRECTED DATA TYPES.**
- **RENAMED COLUMNS.**
- **REMOVED UNNECESSARY FIELDS.**
- **HANDLED NULL VALUES.**
- **STANDARDIZED CATEGORY NAMES.**
- **CREATED MONTH NAME COLUMNS.**
- **CREATED YEAR HIERARCHY.**
- **OPTIMIZED DATA MODEL RELATIONSHIPS.**



DAX MEASURES & BUSINESS CALCULATIONS :-

OVERVIEW

DAX (DATA ANALYSIS EXPRESSIONS) WAS USED TO CREATE DYNAMIC MEASURES THAT CALCULATE KEY BUSINESS METRICS ACROSS SALES, CUSTOMER, CAMPAIGN, AND INVENTORY DASHBOARDS. THESE MEASURES ENABLE REAL-TIME KPI CALCULATIONS AND INTERACTIVE REPORTING BASED ON USER SELECTIONS.

KEY DAX MEASURES

MEASURE	PURPOSE
---------	---------

1. TOTAL SALES :-

PURPOSE :- CALCULATES THE TOTAL REVENUE GENERATED FROM ALL COMPLETED SALES TRANSACTIONS.

DAX FORMULA :-

TOTAL SALES = SUM(FACT_SALES[SALES_AMOUNT])

BUSINESS USE :- THIS MEASURE SERVES AS THE PRIMARY KPI FOR EVALUATING OVERALL BUSINESS PERFORMANCE AND IS USED ACROSS MULTIPLE DASHBOARDS.

2. TOTAL CUSTOMERS :-

PURPOSE :- COUNTS THE TOTAL NUMBER OF UNIQUE CUSTOMERS WHO MADE PURCHASES.

DAX FORMULA :-

TOTAL CUSTOMERS = DISTINCTCOUNT(DIM_CUSTOMER[CUSTOMER_KEY])

BUSINESS USE :- HELPS EVALUATE CUSTOMER GROWTH AND SUPPORTS CUSTOMER SEGMENTATION ANALYSIS.

3. TOTAL ORDERS :-

PURPOSE :- CALCULATES THE TOTAL NUMBER OF CUSTOMER ORDERS.

DAX FORMULA :-

TOTAL ORDERS = COUNTROWS(FACT_SALES)

BUSINESS USE :- MEASURES OVERALL BUSINESS ACTIVITY AND PURCHASING VOLUME.

4. AVERAGE REVENUE PER CUSTOMER :-

PURPOSE :- CALCULATES THE AVERAGE REVENUE GENERATED BY EACH CUSTOMER.

DAX FORMULA :-

**AVERAGE REVENUE PER CUSTOMER = DIVIDE([TOTAL SALES],
[TOTAL CUSTOMERS])**

BUSINESS USE :- IDENTIFIES CUSTOMER VALUE AND SUPPORTS CUSTOMER PROFITABILITY ANALYSIS.

5. AVERAGE ORDERS PER CUSTOMER :-

PURPOSE :- CALCULATES THE AVERAGE NUMBER OF ORDERS PLACED BY EACH CUSTOMER.

DAX FORMULA :-

**AVERAGE ORDERS PER CUSTOMER = DIVIDE([TOTAL ORDERS],
[TOTAL CUSTOMERS])**

BUSINESS USE :- MEASURES CUSTOMER PURCHASING FREQUENCY AND ENGAGEMENT.

6. CAMPAIGN REVENUE :-

PURPOSE :- CALCULATES TOTAL REVENUE GENERATED THROUGH MARKETING CAMPAIGNS.

DAX FORMULA :-

CAMPAIGN REVENUE = [TOTAL SALES]

BUSINESS USE :- EVALUATES THE FINANCIAL PERFORMANCE OF MARKETING CAMPAIGNS.

7. CAMPAIGN SPEND :-

PURPOSE :- CALCULATES THE TOTAL MARKETING EXPENDITURE.

DAX FORMULA :-

CAMPAIGN SPEND = SUM(FACT_CAMPAIGN_SPEND[SPEND])

BUSINESS USE :- TRACKS CAMPAIGN INVESTMENT AND MARKETING COSTS.

8. RETURN ON INVESTMENT (ROI) :-

PURPOSE :- MEASURES THE PROFITABILITY OF MARKETING CAMPAIGNS BY COMPARING REVENUE WITH CAMPAIGN EXPENDITURE.

DAX FORMULA :-

ROI = DIVIDE([CAMPAIGN REVENUE] - [CAMPAIGN SPEND], [CAMPAIGN SPEND], 0)

BUSINESS USE :- HELPS IDENTIFY WHICH CAMPAIGNS GENERATE THE HIGHEST RETURN AND WHICH REQUIRE OPTIMIZATION.

9. CLICK-THROUGH RATE (CTR) :-

PURPOSE :- MEASURES THE EFFECTIVENESS OF MARKETING CAMPAIGNS BY CALCULATING THE PERCENTAGE OF USERS WHO CLICKED ON ADVERTISEMENTS.

DAX FORMULA :-

**CTR = DIVIDE(SUM(FACT_CAMPAIGN_SPEND[CLICKS]),
SUM(FACT_CAMPAIGN_SPEND[IMPRESSIONS]), 0)**

BUSINESS USE :- EVALUATES AUDIENCE ENGAGEMENT AND CAMPAIGN EFFECTIVENESS.

10. TOTAL INVENTORY UNITS :-

PURPOSE :- CALCULATES THE TOTAL QUANTITY OF PRODUCTS AVAILABLE IN INVENTORY.

DAX FORMULA :- TOTAL INVENTORY UNITS = SUM(FACT_INVENTORY[UNITS])

BUSINESS USE :- SUPPORTS INVENTORY MONITORING AND STOCK AVAILABILITY ANALYSIS.

11. INVENTORY VALUE :-

PURPOSE :- CALCULATES THE TOTAL MONETARY VALUE OF THE AVAILABLE INVENTORY.

DAX FORMULA :-

**INVENTORY VALUE = SUMX(FACT_INVENTORY,
FACT_INVENTORY[UNITS] * RELATED(DIM_PRODUCT[PRICE]))**

BUSINESS USE :- HELPS MANAGEMENT UNDERSTAND THE FINANCIAL VALUE OF INVENTORY ASSETS.

12. TOTAL PRODUCTS :-

PURPOSE :- COUNTS THE NUMBER OF UNIQUE PRODUCTS AVAILABLE.

DAX FORMULA :-

TOTAL PRODUCTS = DISTINCTCOUNT(DIM_PRODUCT[PRODUCT_KEY])

BUSINESS USE :- PROVIDES AN OVERVIEW OF PRODUCT DIVERSITY AND INVENTORY COVERAGE.

13. TOTAL CATEGORIES :-

PURPOSE :- CALCULATES THE NUMBER OF UNIQUE PRODUCT CATEGORIES.

DAX FORMULA :-

TOTAL CATEGORIES = DISTINCTCOUNT(DIM_PRODUCT[CATEGORY])

BUSINESS USE:- USED TO ANALYZE PRODUCT DISTRIBUTION ACROSS BUSINESS CATEGORIES.

14. AVERAGE UNITS PER PRODUCT :-

PURPOSE :- CALCULATES THE AVERAGE INVENTORY QUANTITY AVAILABLE PER PRODUCT.

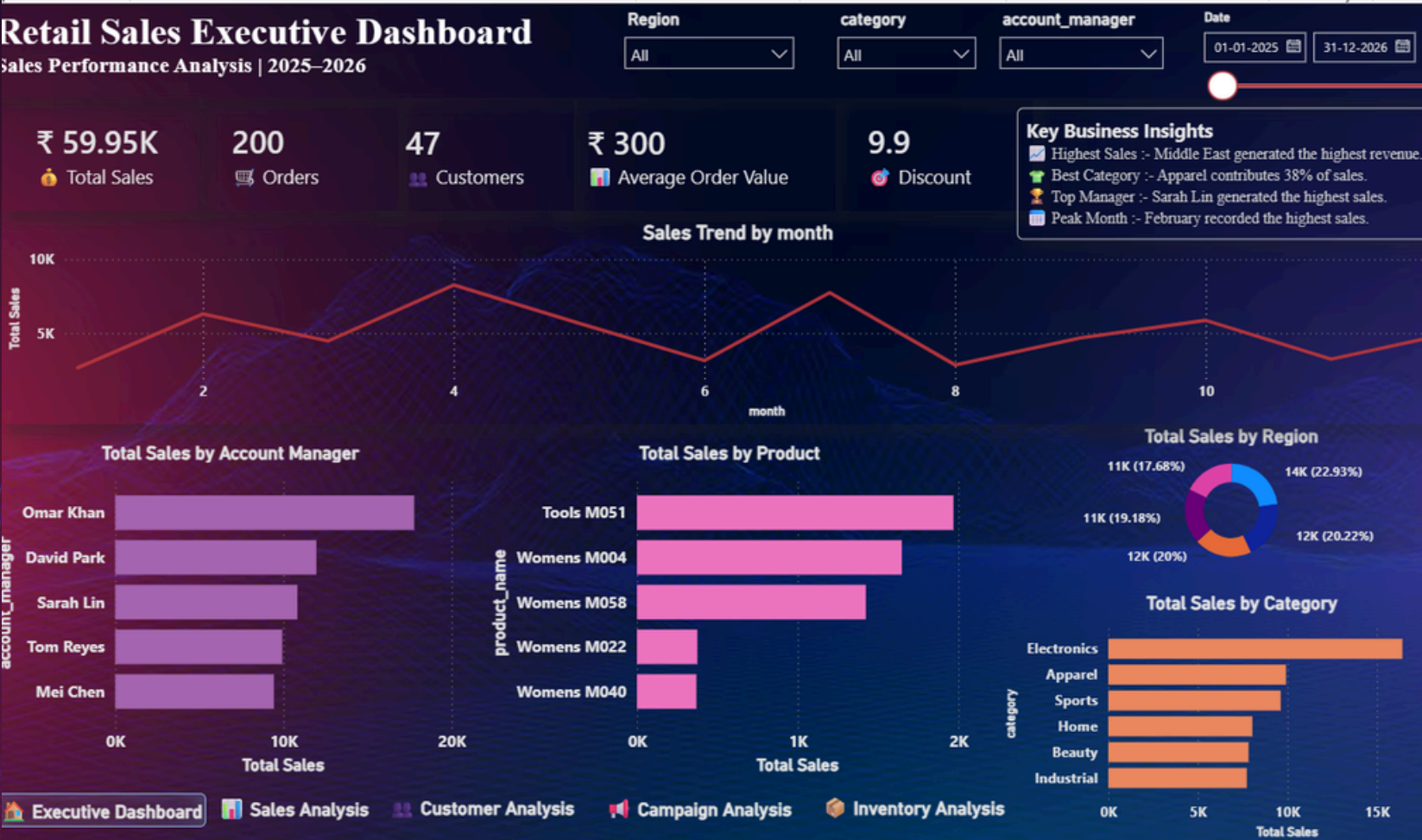
DAX FORMULA :-

**AVERAGE UNITS PER PRODUCT = DIVIDE([TOTAL INVENTORY UNITS],
[TOTAL PRODUCTS])**

BUSINESS USE :- SUPPORTS INVENTORY PLANNING AND STOCK MANAGEMENT.



EXECUTIVE DASHBOARD :-

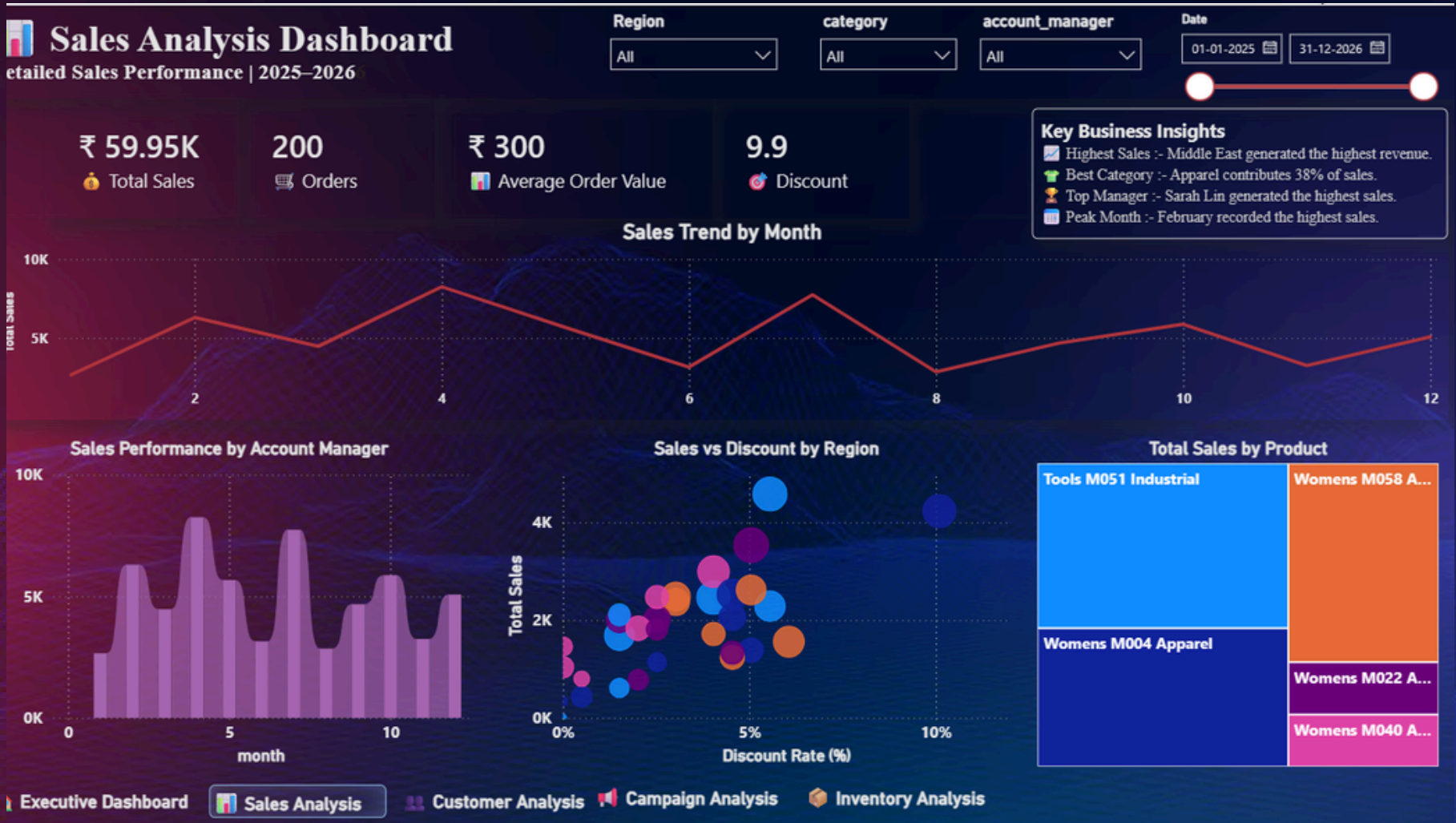


DASHBOARD DESCRIPTION :-THE EXECUTIVE DASHBOARD PROVIDES AN OVERALL SUMMARY OF BUSINESS PERFORMANCE BY DISPLAYING KPIS, REVENUE TRENDS, CUSTOMER METRICS, AND REGIONAL PERFORMANCE.

- KEY VISUALS**
- KPI CARDS
 - LINE CHART
 - DONUT CHART
 - BAR CHART
 - MATRIX
 - BUSINESS INSIGHTS



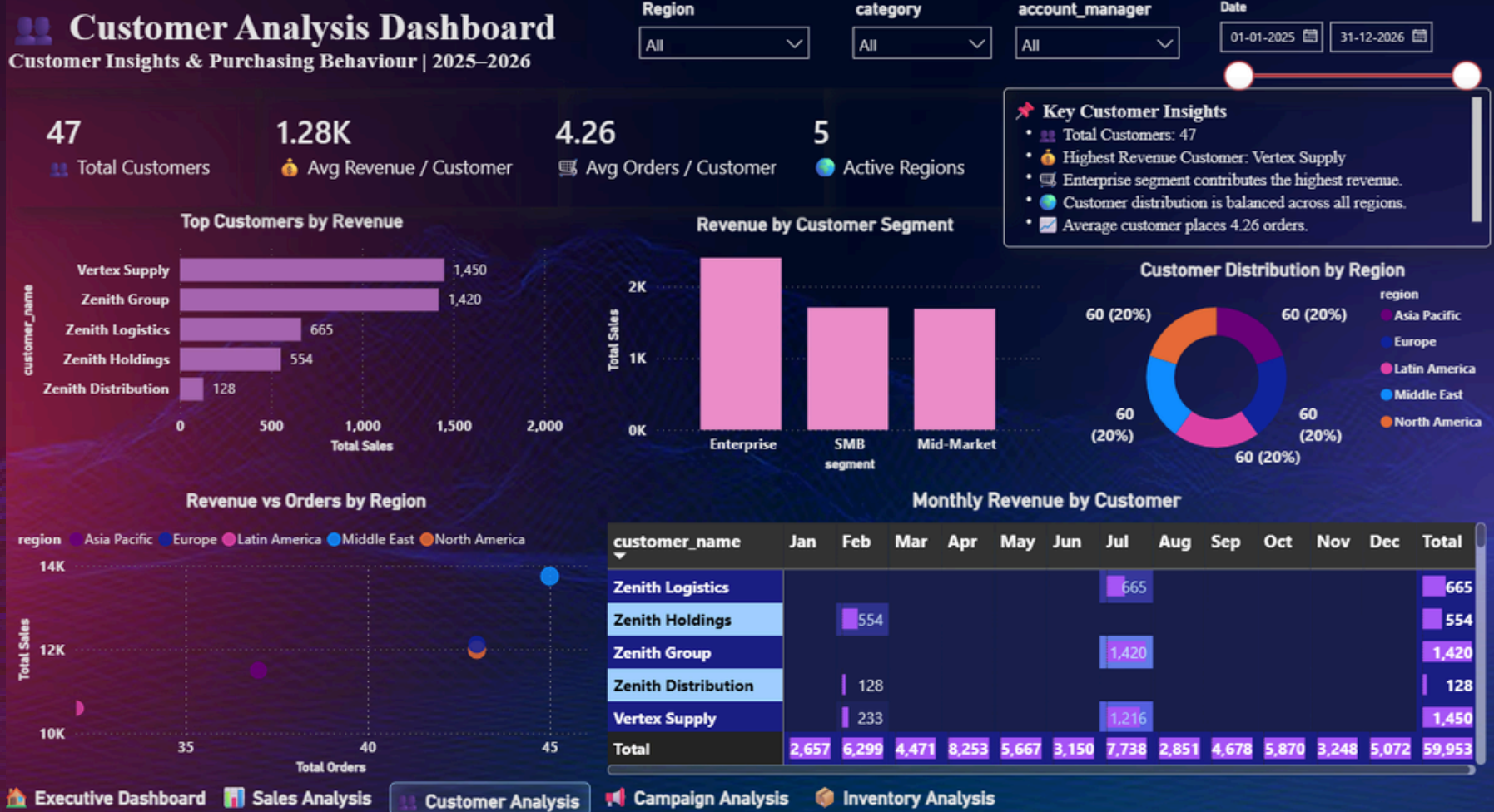
SALES DASHBOARD :-



EXPLAIN:

- REVENUE BY PRODUCT
- REVENUE BY CATEGORY
- REGIONAL SALES
- MONTHLY TRENDS
- PRODUCT PERFORMANCE

CUSTOMER DASHBOARD :-

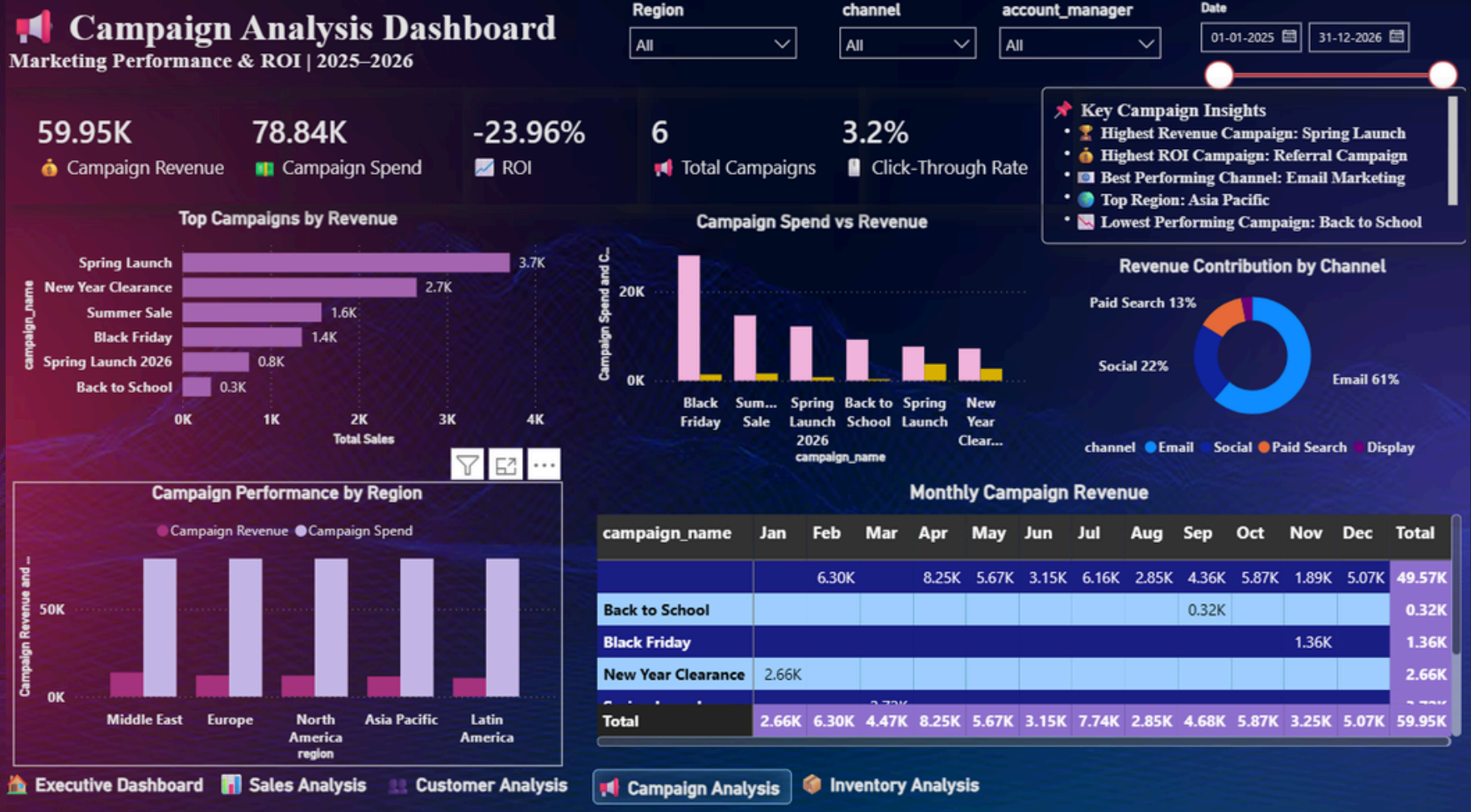


EXPLAIN: -

- CUSTOMER SEGMENTATION
- TOP CUSTOMERS
- REVENUE DISTRIBUTION
- MONTHLY CUSTOMER REVENUE
- CUSTOMER INSIGHTS



CAMPAIGN DASHBOARD :-

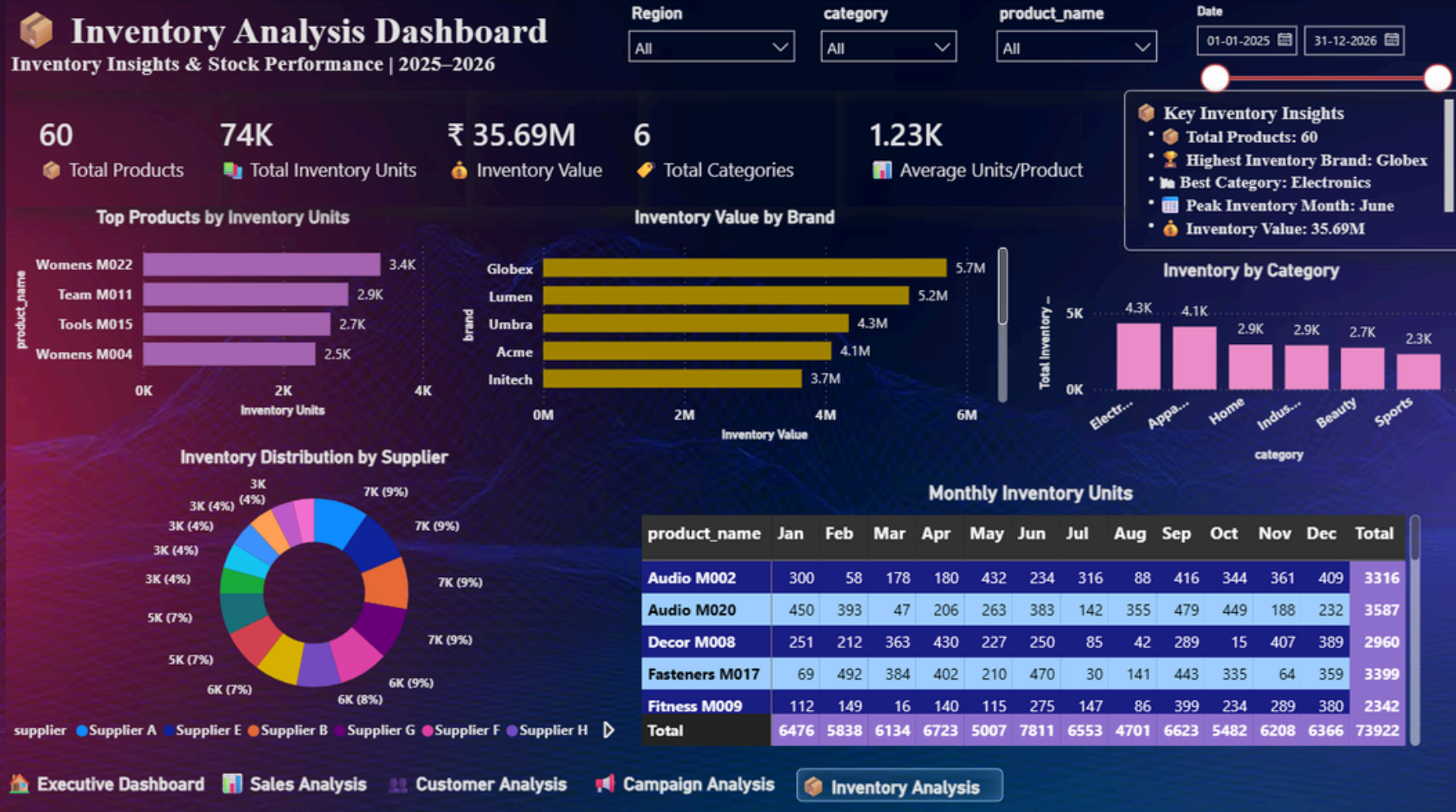


EXPLAIN:-

- CAMPAIGN REVENUE
- CAMPAIGN SPEND
- ROI
- CTR
- CHANNEL PERFORMANCE
- CAMPAIGN COMPARISON



INVENTORY DASHBOARD :-



EXPLAIN:-

- INVENTORY UNITS
- INVENTORY VALUE
- CATEGORY PERFORMANCE
- BRAND PERFORMANCE
- SUPPLIER DISTRIBUTION
- MONTHLY INVENTORY



BUSINESS INSIGHTS :-

SUMMARIZE THE MOST IMPORTANT FINDINGS.

EXAMPLE:

- **TOP PRODUCTS GENERATED THE MAJORITY OF REVENUE.**
 - **ENTERPRISE CUSTOMERS CONTRIBUTED THE HIGHEST SALES.**
 - **EMAIL CAMPAIGNS ACHIEVED THE STRONGEST MARKETING PERFORMANCE.**
 - **INVENTORY VALUE IS CONCENTRATED AMONG A FEW BRANDS.**
 - **CERTAIN REGIONS CONSISTENTLY OUTPERFORM OTHERS.**
- (USE INSIGHTS THAT MATCH YOUR ACTUAL DATA.)**



CHALLENGES FACED :-

EXPLAIN HONESTLY WHAT YOU ENCOUNTERED.

EXAMPLES:

- **DESIGNING AN OPTIMIZED STAR SCHEMA.**
- **CREATING DAX MEASURES.**
- **HANDLING DUPLICATE RECORDS.**
- **BUILDING RELATIONSHIPS.**
- **CHOOSING EFFECTIVE VISUALIZATIONS.**
- **MAINTAINING A CONSISTENT DASHBOARD THEME.**



SKILLS DEMONSTRATED :-

LIST THE SKILLS YOU APPLIED.

TECHNICAL SKILLS

- POWER BI
- SQL
- DAX
- POWER QUERY
- DATA MODELING
- DATA VISUALIZATION
- DASHBOARD DESIGN

ANALYTICAL SKILLS

- KPI ANALYSIS
- TREND ANALYSIS
- BUSINESS STORYTELLING
- DATA CLEANING
- REPORT DEVELOPMENT

FUTURE IMPROVEMENTS :-

EXAMPLES:

- **FORECASTING**
- **DRILL-THROUGH PAGES**
- **ROW-LEVEL SECURITY**
- **POWER BI SERVICE DEPLOYMENT**
- **INCREMENTAL REFRESH**
- **AI VISUALS**
- **REAL-TIME REPORTING**

CONCLUSION :-

THIS PROJECT DEMONSTRATES THE COMPLETE BUSINESS INTELLIGENCE DEVELOPMENT LIFECYCLE, FROM DATA PREPARATION AND MODELING TO INTERACTIVE DASHBOARD CREATION AND BUSINESS INSIGHT GENERATION. BY INTEGRATING MULTIPLE BUSINESS DOMAINS INTO A UNIFIED REPORTING SOLUTION, THE DASHBOARDS ENABLE STAKEHOLDERS TO MAKE INFORMED, DATA-DRIVEN DECISIONS. THE PROJECT SHOWCASES PRACTICAL SKILLS IN POWER BI, SQL, DAX, AND DATA VISUALIZATION, MAKING IT A STRONG PORTFOLIO PROJECT FOR DATA ANALYST ROLES



REFERENCES & CONTACT :-

REFERENCES

- MICROSOFT POWER BI DOCUMENTATION
- MICROSOFT LEARN
- DAX DOCUMENTATION

AUTHOR

MUSKAN SAINI

EMAIL: MK6066863@GMAIL.COM

LINKEDIN: [WWW.LINKEDIN.COM/IN/MUSKAN-SAINI006](https://www.linkedin.com/in/muskan-saini006)

GITHUB: [HTTPS://GITHUB.COM/MUSKAN-SAINI27](https://github.com/muskan-saini27)